

MONSIJ BISWAL

Electronics and Communication Engineering
National Institute of Technology, Durgapur

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EDUCATION

- **National Institute of Technology** Durgapur, India
Electronics and Communication Engineering : GPA: 9.53 July. 2016 – Present
- **Kendriya Vidyalaya** Kharagpur, India
Computer Science; Percentage: 93.80% July 2004 – June. 2016

CAREER OBJECTIVE

I am a third-year undergraduate student with an objective to pursue research and development opportunities in networks and communications focusing on solutions to real-world problems.

RESEARCH INTERESTS

Analog and Digital Communication Systems, Channel Coding, Distributed Storage Systems

EXPERIENCE

- **Indian Institute of Technology** Kharagpur, India
Summer Intern May 2018 - June 2018
 - **Repair-Bandwidth Tradeoff:** There is a trade off between the number of nodes used to recover a damaged node and the amount of data packets stored in a node. The two extreme points in this trade off lead to minimum storage regeneration(MSR) and minimum bandwidth regeneration(MBR)
 - **Minimum Bandwidth Regeneration:** Regenerating codes are a type of distributed storage codes which help a network to store information across several nodes and at the same time possess the ability to recover a failed node

PROJECTS

- **Transferro : LAN-based File Transfer System:** This Python-based application is designed for instantly transferring files between devices connected over the same network when the internet speed is low.
- **Number Guessing Game (MATLAB):** The GUI-based-program chooses a random number that has to be guessed by the player based on the hints generated.
- **CJudge Automate:** A python script involving Selenium and Splinter package to automate submissions of programs to online judges.
- **Minimalist Audio Player(MATLAB):** A simple GUI-based audio player with the support for most common audio formats.

PROGRAMMING SKILLS

- **Languages:** C, C++, Python, Prolog, Lisp, Verilog
- **Softwares:** Matlab, Icarus Verilog, LaTeX, Xilinx ISE

ACADEMIC ACHIEVEMENTS

- Currently ranked among top 5 in the institute.
- Awarded distinction for securing an All India Rank of 19 in KVS Mathematics Olympiad. Selected for INMO training camp.
- Secured a state rank of 4 in SOF National Cyber Olympiad, 2014.
- Represented Kendriya Vidyalaya in Wiz National Spell Bee State Championship, 2015.

EXTRA CURRICULAR ACTIVITIES

- Project titled "Advanced Vehicle Management System" selected for presentation at Diversion 2018, ACM IEM Chapter Kolkata
- Executive member of IEEE Student Branch - NIT Durgapur
- Training and Placement Representative
- Achieved a global rank of 4 in Code Garage September Challenge
- Secured a global rank of 112 in Codechef December Long Coding Contest
- Represented my school at KVS National Youth Parliament

COURSES COMPLETED

Digital Electronics
Analog Communications
Digital signal processing
Network synthesis and analysis

Signals and systems
Analog Electronics
Electromagnetic Theory
Digital communication